

Quiz 5

1 Grading Scheme

- Question 1 is for 1 point (0.5 for correct answer and 0.5 for justification)
- Question 2 is for 3 points (1 point for each justification)
- Question 3 is for 4 points (1 point each for reflexive and symmetric and 2 points for transitive)

Total grade is 8.

2 Solutions

1. $10!/2!213!$. There are $10!$ ways to arrange 10 distinct letters. In the word bookkeeper, 'o' repeats twice, 'k' repeats twice and 'e' repeats thrice.
2.
 - It is reflexive as for all $x \in \mathbb{Z}, x \leq x$.
 - It is not symmetric. Let $x = 1$ and $y = 2$. Then $x \leq y$ but $y \not\leq x$.
 - It is transitive. Let $x, y, z \in \mathbb{Z}$ be integers such that $x \leq y$ and $y \leq z$. Then we have $x \leq z$.
3.
 - Reflexive: $x \equiv x \pmod{n}$ as $x - x = 0$ and $n|0$.
 - Symmetric: Let $x \equiv y \pmod{n}$. By definition $n|(x - y)$. That is there exist $c \in \mathbb{Z}$, such that $c \times n = x - y \pmod{n}$. Then we have $n \times (-c) = y - x$. That is, $n|(y - x)$. Therefore, $y \equiv x \pmod{n}$.
 - Transitivity: Suppose $x \equiv y \pmod{n}$ and $y \equiv z \pmod{n}$. By definition, $n|(x - y)$ and $n|(y - z)$. That is, there exist $c_1, c_2 \in \mathbb{Z}$ such that $n \times c_1 = (x - y)$ and $n \times c_2 = (y - z)$. Adding the equations, we get $n \times (c_1 + c_2) = (x - y) + (y - z) = (x - z)$. That is, $n|(x - z)$. Hence, $x \equiv z \pmod{n}$.

Therefore, $(\text{mod } n)$ is an equivalence relation.